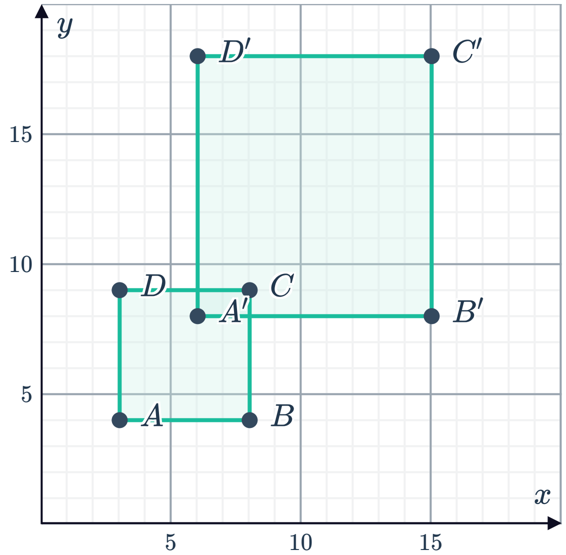
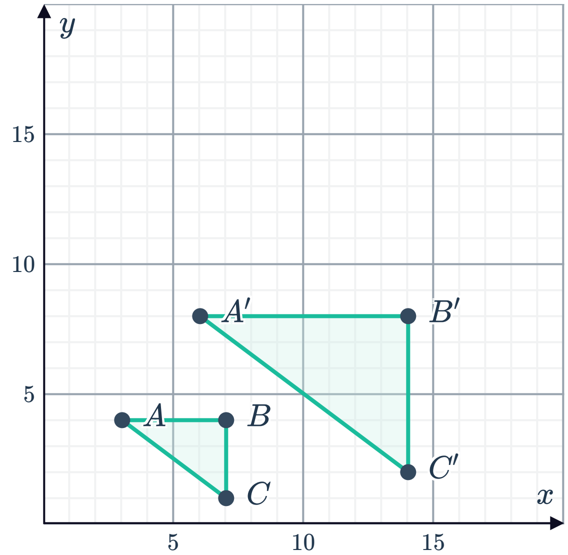
# Dilations

1 Determine whether the following graphs show a dilation. If yes, find the scale factor.

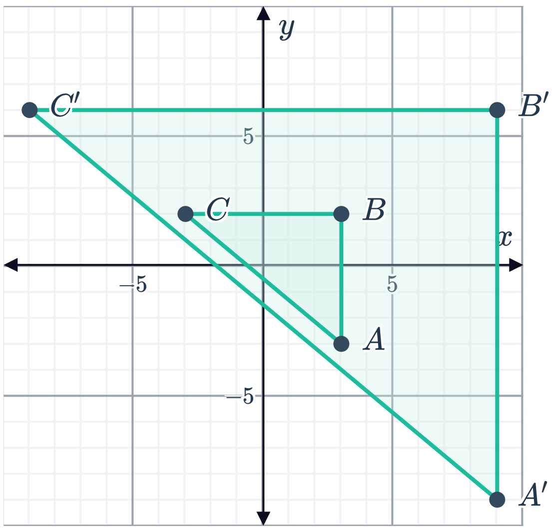
a



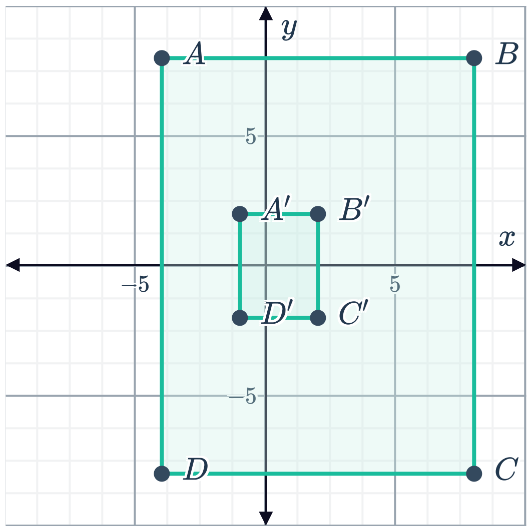

b




c

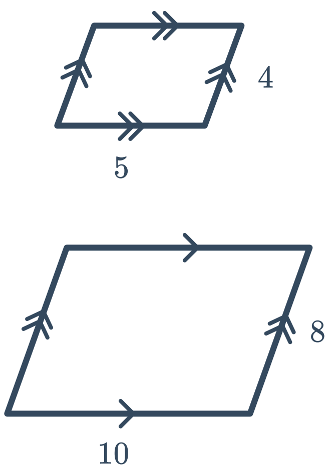



d

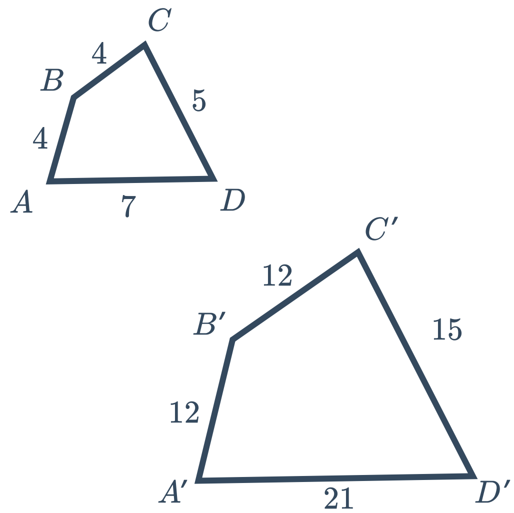


**2** Find the scale factor for each pair of similar [redacted]es:

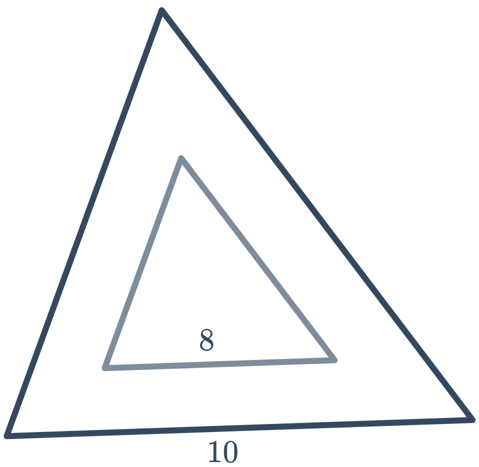
**a**



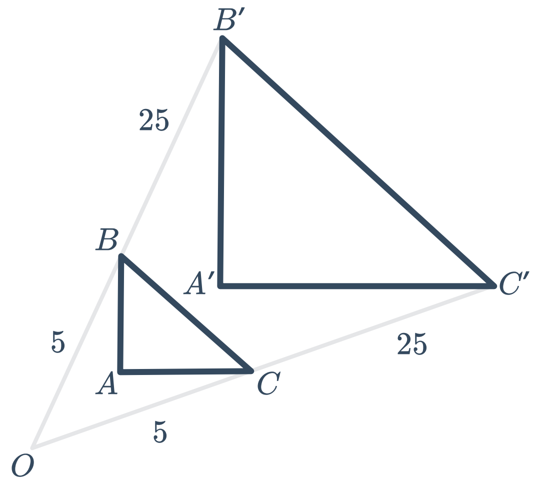

**b**




**c**

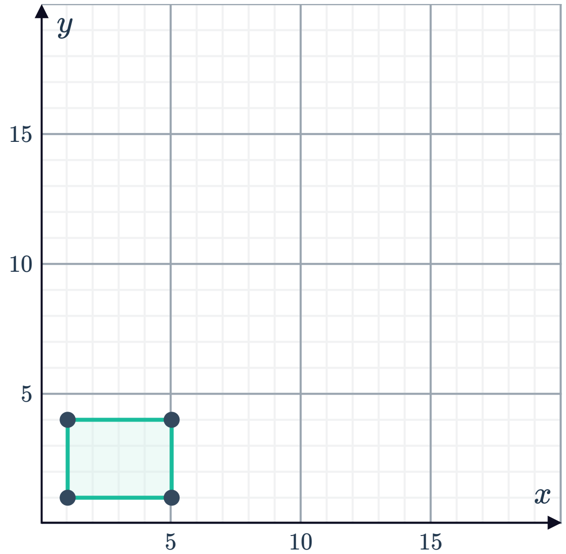



**d**

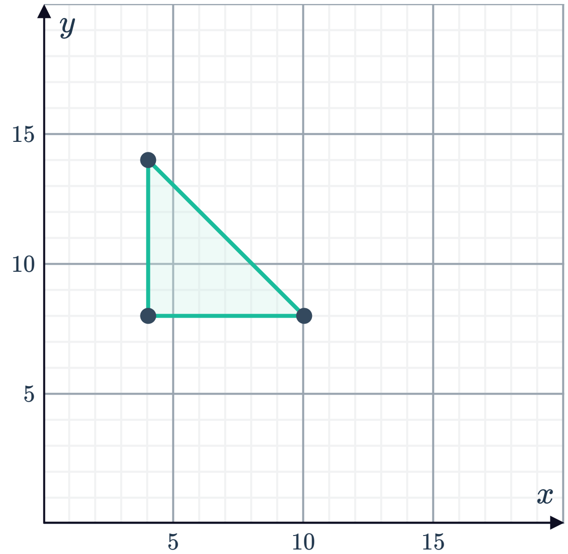


**3** Dilate the figure by the given factor using th[redacted]gin as the center of dilation:

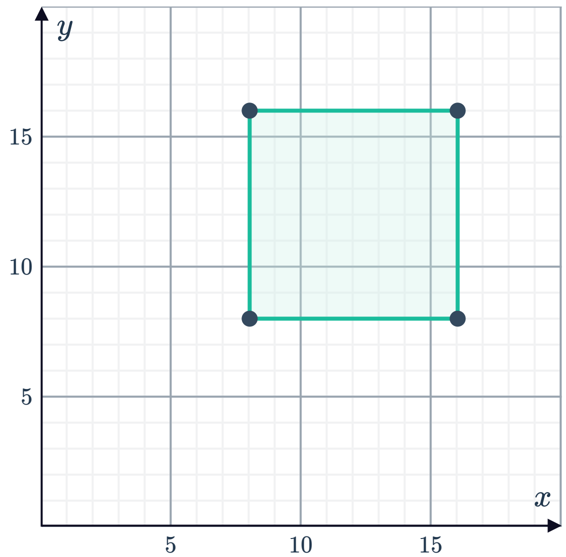
**a** Scale factor of $3$



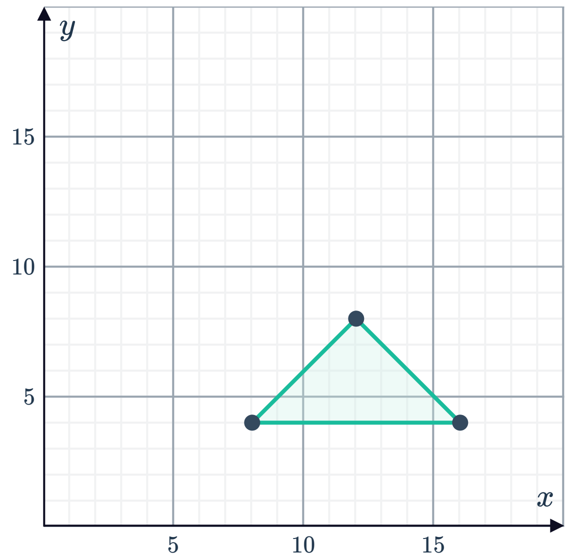

**b** Scale factor of $4$




**c** Scale factor of $\frac{1}{2}$



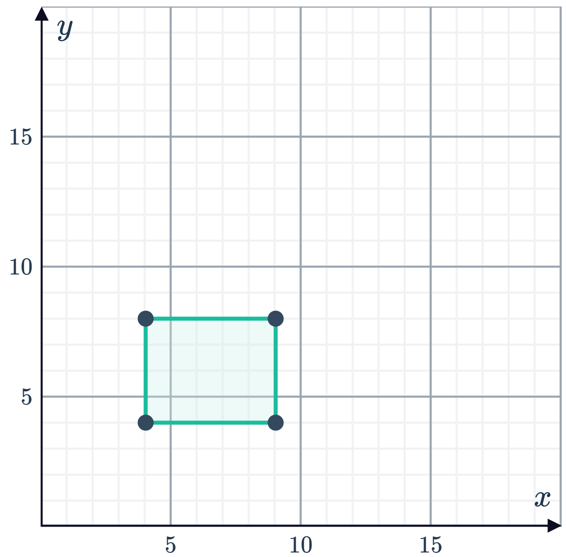

**d** Scale factor of $\frac{5}{4}$



**4** Dilate each figure using the given scale fact[redacted]d center of dilation.

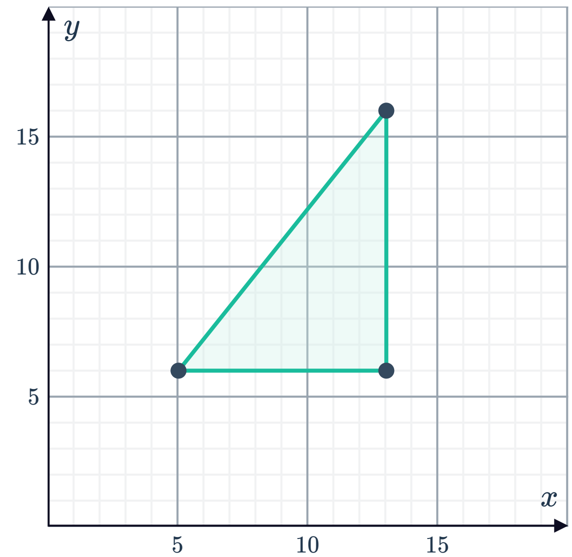
**a** Scale factor: $2$

Center of dilation: $A\left(2,3\right)$




**b** Scale factor: $\frac{1}{2}$

Center of dilation: $A\left(3,2\right)$




**5** A triangle with vertices $A\left(-7,-5\right)$, $B\left(3,-5\right)$ and $C\left(-5,9\right)$ is dilated by a factor of $0.5$ using the origin as the center of dilation. What are the coordinates of the vertices of the dilated triangle?

**6** A rectangle with vertices $P\left(-5,3\right)$, $Q\left(3,3\right)$, $R\left(3,-5\right)$ and $R\left(-5,-5\right)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $P'\left(-25,15\right)$, $Q'\left(15,15\right)$ , $R'\left(15,-25\right)$ and $S'\left(-25,-25\right)$.

Find the scale factor.